Package Insert

1. Product Name

Gabenil 300 Capsules

2. Name and Strength of the Active Ingredient

Each Gabenil 300 mg capsule contains 300 mg Gabapentin.

3. Product Description

Hard gelatin yellow/yellow capsules of size 1, containing gabapentin in white to off-white crystalline powder.

4. Pharmacological Properties

The precise mechanism of action of gabapentin is not known.

Gabapentin is structurally related to the neurotransmitter GABA (gamma-aminobutyric acid) but its mechanism of action is different from that of several other active substances that interact with GABA synapses, including valproate, barbiturates, benzodiazepines, GABA transaminase inhibitors, GABA uptake inhibitors, GABA agonists, and GABA prodrugs. *In vitro* studies with radiolabeled gabapentin have characterized a novel peptide binding site in rat brain tissues including neocortex and hippocampus that may relate to anticonvulsant and analgesic activity of gabapentin and its structural derivatives.

The binding site for gabapentin has been identified as the alpha$_2$-delta subunit of voltage-gated calcium channels.

Pharmacodynamics

Pharmacotherapeutic groups: Other antiepileptics ATC code: N03AX12

Gabapentin at relevant clinical concentrations does not bind to other common drug or neurotransmitter receptors of the brain including $GABA_A$, $GABA_B$, benzodiazepine, glutamate, glycine or N-methyl-d-aspartate receptors.

Gabapentin does not interact with sodium channels *in vitro* and so differs from phenytoin and carbamazepine. Gabapentin partially reduces responses to the glutamate agonist N-methyl-D-aspartate (NMDA) in some test systems *in vitro*, but only at concentrations greater than 100 μM, which are not achieved *in vivo*. Gabapentin administration to rats increases GABA turnover in several brain regions in a manner similar to valproate sodium, although in different regions of brain. The relevance of these various actions of gabapentin to the anticonvulsant effects remains to be established. In animals, gabapentin readily enters the brain and prevents seizures from maximal electroshock, from chemical convulsants including inhibitors of GABA synthesis, and in genetic models of seizures.

Pharmacokinetics

Absorption
Following oral administration, peak plasma gabapentin concentrations are observed within 2 to 3 hours. Gabapentin bioavailability (fraction of dose absorbed) tends to decrease with increasing dose. Absolute bioavailability of a 300 mg capsule is approximately 60%. Food, including a high-fat diet, has no clinically significant effect of gabapentin pharmacokinetics.

Gabapentin pharmacokinetics are not affected by repeated administration. Although plasma gabapentin concentrations were generally between 2 µg/ml and 20 µg/ml in clinical studies, such concentrations were not predictive of safety or efficacy.
Pharmacokinetic parameters are given in Table 3.

Pharmacokinetic Parameter	300 mg (N = 7)		400 mg (N = 14)		800 mg (N = 14)	
	Mean	%CV	Mean	%CV	Mean	%CV
C_{max} (µg/ml)	4.02	(24)	5.74	(38)	8.71	(29)
T_{max} (hr)	2.7	(18)	2.1	(54)	1.6	(76)
T1/2 (hr)	5.2	(12)	10.8	(89)	10.6	(41)
AUC (0-8) µg·hr/ml	24.8	(24)	34.5	(34)	51.4	(27)
Ae% (%)	NA	NA	47.2	(25)	34.4	(37)

C_{max} = Maximum steady state plasma concentration
T_{max} = Time for C_{max}
T1/2 = Elimination half-life
AUC (0-8) = Steady state area under plasma concentration-time curve from time 0 to 8 hours post dose
Ae% = Percent of dose excreted unchanged into the urine from time 0 to 8 hours postdose
NA = Not available

Distribution
Gabapentin is not bound to plasma protein and has a volume of distribution equal to 57.7 litres. In patients with epilepsy, gabapentin concentrations in cerebrospinal fluid (CSF) are approximately 20% of corresponding steady-state through plasma concentrations. Gabapentin is present in the breast milk of breast-feeding women.

Metabolism
There is no evidence of gabapentin metabolism in humans. Gabapentin does not include hepatic mixed function oxidase enzymes responsible for drug metabolism.

Elimination
Gabapentin is eliminated unchanged solely by renal function. The elimination half-life is independent of dose and averages of 5 to 7 hours.

In elderly patients, and in patients with impaired renal function, gabapentin plasma clearance is reduced. Gabapentin elimination-rate constant, plasma clearance, and renal clearance are directly proportional to creatinine clearance.

Gabapentin is removed from plasma by haemodialysis. Dosage adjustment in patients with compromised renal function or undergoing haemodialysis is recommended.

Gabapentin pharmacokinetics in children were determined in 50 healthy subjects between the ages of 1 month and 12 years. In general, plasma gabapentin concentrations in children > 5 years of age are similar to those in adults when dosed on a mg/kg basis.

Linearity / Non-linearity
Gabapentin bioavailability (fraction of dose absorbed) decreases with increasing dose with imparts non-linearity to pharmacokinetic parameters which do not include F such as CLr and T1/2 are best described be linear pharmacokinetics. Steady state plasma gabapentin concentrations are predictable from single dose data.

5. Clinical Particulars

Indications

Epilepsy
Gabapentin is indicated as adjunctive therapy in the treatment of partial seizures with and without secondary generalization in adults and children age 3 years and above. Safety and effectiveness for adjunctive therapy on pediatric patients below the age of 3 years have not been established.

Neuropathic pain
Gabapentin is indicated for the treatment of neuropathic pain which includes diabetic neuropathy, post-herpetic neuralgia and trigeminal neuralgia in adults age 18 years and above. Safety and effectiveness in patients below the age of 18 years have not been established.

Recommended Dosage

Gabapentin is given orally with or without food.
When in the judgement of the clinician there is a need for dose reduction, discontinuation, or substitution with an alternative medication, the dose should be done gradually over a minimum of one week.

Epilepsy

Adults and pediatric patients over 12 years of age:
In clinical trial, the effective dosing range was 900 to 3600 mg/day. Therapy may be initiated by administrating 300 mg three times a day (TID) on Day 1, or by titrating the dose as described in TABLE 1. Thereafter, the dose can be increased in three equally divided doses up to maximum dose of 3600 mg/day. Dosages up to 4800 mg/day have been well tolerated in long term open label clinical studies. The maximum time between doses in the three times a day (TID) schedules should not exceed 12 hours to prevent breakthrough convulsions.

TABLE 1			
DOSAGE CHART – INITIAL TIRTATION			
Dose	Day 1	Day 2	Day 3
900 mg	300 mg QD[a]	300 mg BID[b]	300 mg TID[c]

[a]QD = once a day
[b]BID = two times a day
[c]TID = three times a day

Adults and pediatric patients age 3-12 years:
The starting dose should range from 10-15 mg/kg/day given in equally divided doses (three times a day) and the effective dose reached by upward titration over a period of approximately three days. The effective dose of gabapentin in patients age 5 years and older is 25 to 35 mg/kg/day given in equally divided doses (three times a day). The effective dose in pediatric patients age 3 to less than 5 years is 40 mg/kg/day given in equally divided doses (three times a day). Dosages up to 50 mg/kg/day have been well tolerated in a long term clinical study. The maximum time interval between doses should not exceed 12 hours.

It is not necessary to monitor gabapentin plasma concentrations to optimize gabapentin therapy. Further, gabapentin may be used in combination with other antiepileptic drugs without concern for alteration of the plasma concentrations of gabapentin or serum concentrations of other antiepileptic drugs.

Neuropathic pain in adults
The starting dose is 900 mg/day given as three equally divided doses and increased if necessary, based on response, up to a maximum dose of 3600 mg/day. Therapy should be initiated by titrating the dose as described in TABLE 1.

<u>Dosage adjustment in impaired renal function for patients with neuropathic pain or epilepsy</u>
Dosage adjustment is recommended in patients with compromised renal function as described in TABLE 2 and/or those undergoing hemodialysis.

TABLE 2	
DOSAGE OF GABAPENTIN IN ADULTS BASED ON RENAL FUNCTION	
Creatinine Clearance (mL/min)	Total Daily Dose[a] (mg/day)
≥ 80	900-3600

50-79	600-1800
30-49	300-900
15-29	150[b]-600
< 15	150[b]-300

[a] Total daily dose should be administered as a TID regimen. Doses used to treat patients with normal renal function (creatinine clearance > 80 mL/min) range from 900 to 3600 mg/day. Reduced dosages are for patients with renal impairment (creatinine clearance < 78 mL/min).
[b] To be administered as 300 mg every other day.

Dosage adjustment in patients undergoing hemodialysis
For patients undergoing hemodialysis who have never received gabapentin, a loading dose of 300-400 mg is recommended, then 300 mg of gabapentin following each 4 hours of hemodialysis.

Contraindications

The use of this medicine is contra-indicated to patients with known hypersensitivity to gabapentin or to any of the excipients of the product.

Warnings and Precautions

Although there is no evidence of rebound seizures with Gabenil, the abrupt withdrawal of anticonvulsant agents in epileptic patients may precipitate status epilepticus. Any dose reduction, discontinuation or substitution of alternative anticonvulsant medication should be effected gradually over a period of at least one week.

Gabenil is not generally considered effective in the treatment of absence seizures.

Patients taking Gabenil can be the subject of mood and behavioral disturbances.

Potential for an increase in risk of suicidal thoughts or behaviours.

Special caution is recommended for patients with a history of psychotic episodes.

Patients who require concomitant treatment with morphine may experience increases in gabapentin concentrations. Patients should be carefully observed for signs of CNS depression, such as somnolence, and the dose of gabapentin or morphine should be reduced appropriately.

Important information about some of the ingredients of Gabenil
Gabenil capsules contain lactose (a type of sugar). If you have been told by your doctor that you have an intolerance to some sugars, contact your doctor before taking this medicinal product.

Interactions with other medicines

Concomitant administration with morphine increases mean Gabapentin AUC. Therefore, patients should be carefully observed for signs of CNS depression, such as somnolence, and the dose of gabapentin or morphine should be reduced appropriately.

No interaction between gabapentin and phenobarbital, phenytoin, valproic acid, or carbamazepine has been observed.

Gabapentin steady-state pharmacokinetics are similar for healthy subjects and patients with epilepsy receiving these antiepileptic agents.

Coadministration of gabapentin with oral contraceptives containing norethindrone and/or ethinyl estradiol, does not influence the steady-state pharmacokinetics of either component.

Coadministration of gabapentin with antacids containing aluminium and magnesium reduces gabapentin bioavailability up to 24%. It is recommended that gabapentin be taken at the earliest two hours following antacid administration.

Renal excretion of gabapentin is unaltered by probenecid.

A slight decrease in renal excretion of gabapentin that is observed when it is coadministered with cimetidine is not expected to be of clinical importance.

Pregnancy and Lactation

Safe use in humans has not been established. Therefore, Gabenil should be avoided during pregnancy.
Gabapentin is excreted into human milk and therefore it is not recommended during lactation.

Side Effects

The adverse reactions observed during clinical studies conducted in epilepsy (adjunctive and monotherapy) and neuropathic pain have been provided in a single list below by class and frequency (very common (≥ 1/10); common (≥ 1/100 to <1/10); uncommon (≥1/1000 to <1/100); rare (≥1/10000 to <1/100); very rare (<1/10000). Where an adverse reaction was seen at different frequencies in clinical studies, it was assigned to the highest frequency reported.

Additional reactions reported from post-marketing experience are included as frequency Not known (cannot be estimated from the available data) in italics in the list below.

Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

Body system	**Adverse drug reactions**
Infections and infestations	

Very Common	Viral infection
Common	Pneumonia, respiratory infection, urinary tract infection, infection, otitis media
Blood and the lymphatic system disorders	
Common	Leucopenia
Not Known	***Thrombocytopenia***
Immune system disorders	
Uncommon	Anorexia, increased appetite
Psychiatric disorders	
Common	Hostility, confusion and emotional lability, depression, anxiety, nervousness, thinking abnormal.
Not Known	***Hallucinations***
Nervous system disorders	
Very Common	Somnolence, dizziness, ataxia
Common	Convulsions, hyperkinesias, dysarthria, amnesia, tremor, insomnia, headache, sensations such as paresthesia, hypaesthesia, coordination abnormal, nystagmous, increased, decreased or absent reflexes
Uncommon	Hypokinesia
Not Known	*Other movement disorders (e.g. choreoathetosis, dyskinesia, dystonia)*
Eye disorders	
Common	Visual disturbances such as amblyopia, diplopia
Ear and Labyrinth disorders	
Common	Vertigo
Not Known	*Tinnitus*
Cardiac disorders	
Uncommon	Palpitations
Vascular disorders	
Common	Hypertension, vasodilation
Respiratory, thoracic and mediastinal disorders	
Common	Dyspnoea, bronchitis, pharyngitis, cough, rhinitis
Gastrointestinal disorders	
Common	Vomiting, nausea, dental abnormalities, gingivitis, diarrhea, abdominal pain, dyspepsia, constipation, dry mouth or throat, flatulence
Not Known	***Pancreatitis***
Hepatobiliary disorders	
Not Known	***Hepatitis, jaundice***

Skin and subcutaneous tissue disorders	
Common	Facial oedema, purpura most often described as bruises resulting from physical trauma, rash, pruritus, acne
Not Known	*Steven-Johnson syndrome, angioedema, erythema multiforme, alopecia*
Musculoskeletal, connective tissue and bone disorders	
Common	Arthralgia, myalgia, back pain, twinching
Not Known	***Myoclonus***
Renal and urinary disorder	
Not Known	***Acute renal failure, incontinence***
Reproductive system and breast disorders	
Common	Impotence
Not Known	***Breast hypertrophy, gynaecomastia***
General disorders and administration site conditions	
Very Common	Fatigue, fever
Common	Peripheral oedema, abnormal gait, asthenia, pain, malaise, flu syndrome

Under treatment with gabapentin cases of acute pancreatitis were reported. Causality with gabapentin is unclear.
In patients on haemodialysis due to end-stage renal failure, myopathy with elevated creatine kinase levels has been reported.
Respiratory tract infections, otitis media, convulsions and bronchitis were reported only in clinical studies in children, aggressive behaviour and hyperkinesias were reported commonly.

Symptoms and Treatment of Overdose

Acute, life-threatening toxicity has not been observed with gabapentin overdoses of up to 49 grams. Symptoms of the overdoses included dizziness, double vision, slurred speech, drowsiness, lethargy and mild diarrhoea. All patients recovered fully with supportive care. Reduced absorption of gabapentin at higher doses may limit drug absorption at the time of overdosing and, hence, minimize toxicity from overdoses.
Although gabapentin can be removed by haemodialysis it is not usually required.
However, in patients with renal impairment, haemodialyisis may be indicated.

6. Pharmaceutical Particulars

Storage Condition

Keep this medicine out of the sight and reach of children.

Do not use this medicine after the expiry date which is stated on the carton label and blister foil. The expiry date refers to the last day of that month.

Store below 30ºC. Protect from light and moisture.

Shelf Life

3 years

MAL number

Gabenil 300: MAL09042396AZ

7. Manufacturer

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8. Date of revision

01/2023

For internal use only: my-pl-gabenil-300mg-400mg-caps-a1t